

Question 1 Random Forest: Full Trait Model

This notebook evaluates the custom CSC 466 random forest using `data/processed/breed_forest_full.csv`.

This version keeps each AKC trait rating as its own predictor, while `Coat Type` and `Coat Length` remain categorical.

Final Test Results

The model is trained on the training split using the best cross-validation parameters, then evaluated on the held-out test split. Macro F1 is useful here because the three popularity tiers are not perfectly balanced.

Model Interpretation

These tables show which features the forest split on most often and how average feature values differ across the actual popularity tiers.

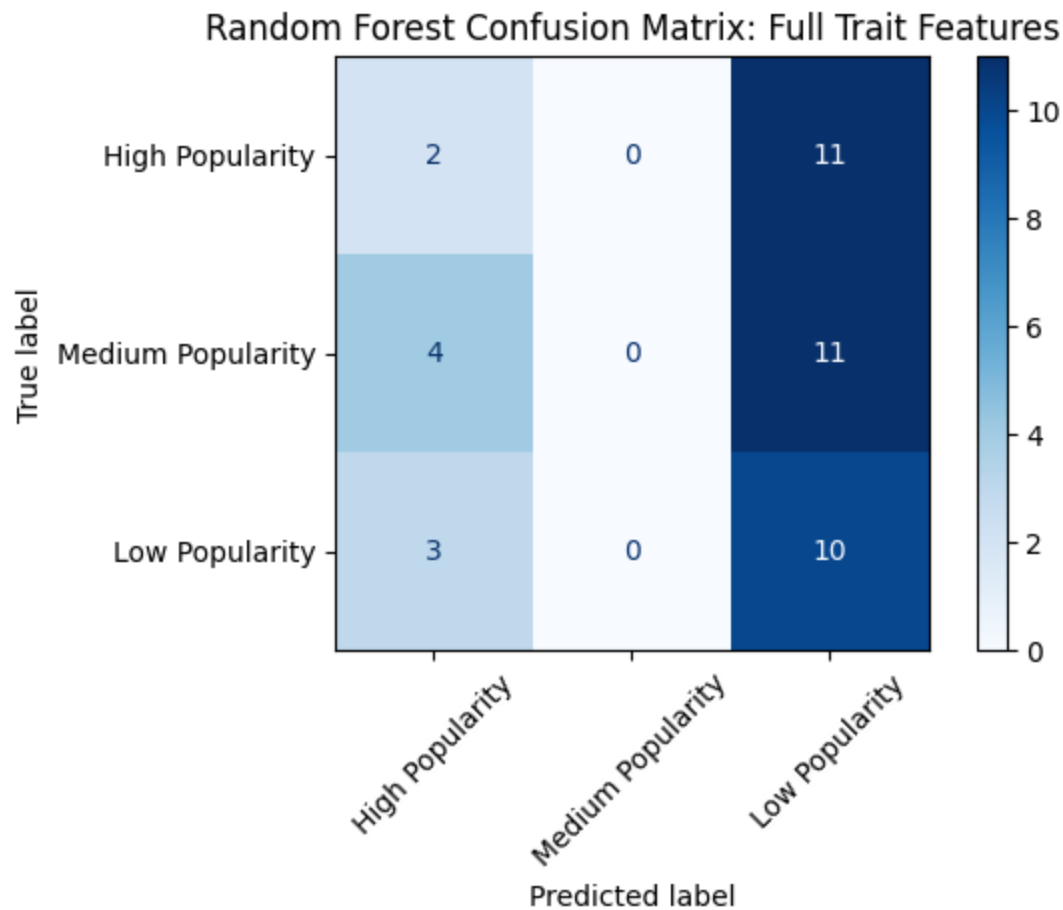
	Feature	Split Count
0	Adaptability Level	18
1	Barking Level	13
2	Watchdog/Protective Nature	12
3	Openness To Strangers	12
4	Good With Other Dogs	11
5	Coat Grooming Frequency	8
6	Energy Level	7
7	Good With Young Children	6
8	Shedding Level	6
9	Playfulness Level	5
10	Trainability Level	5
11	Drooling Level	4
12	Mental Stimulation Needs	4
13	Affectionate With Family	1

Popularity_Tier	High Popularity	Medium Popularity	Low Popularity
Affectionate With Family	4.55	4.51	4.47
Good With Young Children	4.18	3.73	3.71
Good With Other Dogs	3.57	3.45	3.62
Shedding Level	2.74	2.62	2.41
Coat Grooming Frequency	2.42	2.28	2.11
Drooling Level	1.95	1.76	1.61
Openness To Strangers	3.66	3.51	3.29
Playfulness Level	3.88	3.68	3.41
Watchdog/Protective Nature	3.78	3.82	3.55
Adaptability Level	3.92	3.77	3.65
Trainability Level	4.03	3.76	3.80
Energy Level	3.85	3.75	3.65
Barking Level	3.08	3.13	3.23
Mental Stimulation Needs	3.78	3.63	3.61

Popularity_Tier	Coat Type	Coat Length
High Popularity	Smooth	Short
Medium Popularity	Smooth	Short
Low Popularity	Double	Medium

	precision	recall	f1-score	support
High Popularity	0.22	0.15	0.18	13
Medium Popularity	0.00	0.00	0.00	15
Low Popularity	0.31	0.77	0.44	13
accuracy			0.29	41
macro avg	0.18	0.31	0.21	41
weighted avg	0.17	0.29	0.20	41

```
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x14c6cc440>
```



Prediction CSV

This output compares each test-set breed's actual popularity tier to the tier predicted by the full-trait random forest model. **Average Rank** is included for interpretation only; it is not used as a predictor.

	Breed	Average Rank	Actual Popularity Tier	Predicted Popularity Tier	Correct
0	Bloodhound	51.2	High Popularity	Low Popularity	False
1	Dogue de Bordeaux	73.2	Medium Popularity	High Popularity	False
2	Irish Water Spaniel	163.6	Low Popularity	Low Popularity	True
3	Anatolian Shepherd Dog	87.9	Medium Popularity	Low Popularity	False
4	Belgian Malinoi	42.2	High Popularity	Low Popularity	False
5	Italian Greyhound	67.2	Medium Popularity	Low Popularity	False
6	Chinese Shar-Pei	65.5	High Popularity	Low Popularity	False
7	Afghan Hound	108.9	Medium Popularity	Low Popularity	False
8	Polish Lowland Sheepdog	175.6	Low Popularity	High Popularity	False

9	Irish Terrier	124.9	Medium Popularity	Low Popularity	False
10	Pointer	116.0	Medium Popularity	Low Popularity	False
11	Dogo Argentino	87.9	Medium Popularity	High Popularity	False
12	Cane Corso	29.0	High Popularity	Low Popularity	False
13	Belgian Sheepdog	123.8	Medium Popularity	Low Popularity	False
14	Brittany	27.6	High Popularity	Low Popularity	False
15	American English Coonhound	173.1	Low Popularity	Low Popularity	True
16	Neapolitan Mastiff	110.1	Medium Popularity	Low Popularity	False
17	Siberian Husky	17.5	High Popularity	Low Popularity	False
18	Newfoundland	40.4	High Popularity	Low Popularity	False
19	Azawakh	192.4	Low Popularity	Low Popularity	True
20	Spinone Italiano	113.2	Medium Popularity	Low Popularity	False
21	Tibetan Spaniel	119.9	Medium Popularity	Low Popularity	False
22	Shetland Sheepdog	25.3	High Popularity	High Popularity	True
23	Lowchen	161.8	Low Popularity	Low Popularity	True
24	Cirneco dell'Etna	176.8	Low Popularity	Low Popularity	True
25	American Water Spaniel	162.8	Low Popularity	Low Popularity	True
26	English Cocker Spaniel	48.8	High Popularity	Low Popularity	False
27	Bluetick Coonhound	132.5	Medium Popularity	Low Popularity	False
28	Pyrenean Shepherd	182.6	Low Popularity	High Popularity	False
29	Skye Terrier	179.1	Low Popularity	Low Popularity	True
30	Chesapeake Bay Retriever	47.3	High Popularity	Low Popularity	False
31	Cocker Spaniel	29.8	High Popularity	Low Popularity	False
32	Lakeland Terrier	145.5	Low Popularity	Low Popularity	True
33	Maltese	34.8	High Popularity	Low Popularity	False
34	Saint Bernard	52.5	High Popularity	High Popularity	True
35	Canaan Dog	181.8	Low Popularity	Low Popularity	True
36	Irish Red and White Setter	156.6	Low Popularity	High Popularity	False
37	American Hairless Terrier	122.3	Medium Popularity	High Popularity	False
38	Petit Basset Griffon Vendéen	154.0	Low Popularity	Low Popularity	True
39	Keeshond	89.6	Medium Popularity	High Popularity	False
40	Miniature Bull Terrier	113.3	Medium Popularity	Low Popularity	False